Vishal Bharti

Computational Biologist · CRISPR/Cas · RNA Structural Biology · Single-Cell Genomics

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Research Profile

I build computational and statistical methods for genome engineering and RNA biology, currently in the RNA Biology Group at **CSIR-IGIB**. My own work is computational, done alongside wet-lab colleagues. The question I keep returning to is whether a biological conclusion actually follows from the data that produced it, and it runs through most of what I have done: a co-first-author method for predicting intramolecular RNA triple helices (*Biochemistry*, ACS), a single-cell analysis that identified candidate repair genes for staggered versus blunt DNA double-strand breaks, StickForStats, an open-source platform that validates the assumptions behind a statistical test before running it, and a pre-registered audit of whether published biomedical statistics can be recomputed. I came to biology from electronics engineering, taught myself the field over two years, and entered through an M.Tech at IIT Guwahati (GATE-BT **All-India Rank 127, top ~1%**).

Education

M.Tech, Biotechnology · Indian Institute of Technology (IIT) Guwahati · *2020 to 2022* · CGPA 8.75/10
Master's thesis: NMD-susceptible transcripts in UPF3B-knockout cells (RNA-seq).

B.Tech, Electronics & Communication Engineering · Institute of Engineering & Management (IEM), Kolkata · *2014 to 2018* · CGPA 8.19/10

Research Experience

Project Associate-II, RNA Biology Group (Dr. Debojyoti Chakraborty) · CSIR-Institute of Genomics and Integrative Biology (IGIB), New Delhi · **Feb 2023 to present (3+ years)**

- **RNA intramolecular triple-helix prediction (TRIPinRNA).** Co-developed (**co-first author**) a Python method to identify and characterize intramolecular RNA triple helices, applied to X-chromosome-inactivation and gene-regulation contexts; colleagues validated three of its predictions with biophysical assays. *Published in* Biochemistry (*ACS*). Co-supervised by Dr. Souvik Maiti.
- **DNA double-strand-break repair from single-cell transcriptomes.** Analyzed single-cell RNA-seq from a colleague's CRISPRi screen of DNA-repair genes, comparing cells that did and did not accurately repair a reporter cut with staggered (en1FnCas9) or blunt (SpCas9) ends. Worked from count matrices onward in Seurat (QC, integration, differential expression, gene-set comparisons, GO enrichment), producing candidate repair genes for validation; heat map and volcano plots credited in her thesis.
- **Interneuron-migration disorders in human forebrain assembloids.** Led RNA-seq, protein-protein interaction network, and pan-tissue analyses linking engineered forebrain assembloids on polymeric nanofiber scaffolds to interneuron-migration phenotypes. (*Preprint; under revision*, Stem Cell Reports.)
- **MLC leukodystrophy: disease modeling and diagnostics.** Led RNA-seq analysis and a computational approach to detect the 132dupC *MLC1* mutation, supporting iPSC disease-model work on the astrocyte-vacuolation phenotype. (*Published in* Orphanet Journal of Rare Diseases, *2026*.)
- **G-quadruplex regulation of *CCN1*.** Computational analysis and proteomics data analysis of IGF2BP1-G-quadruplex regulation of *CCN1* expression. *Published in* Biochemistry (*ACS*). In collaboration with Dr. Mary K. Ekka.

Data analysis collaborator (remote) · Archaeology PhD thesis, IIT Gandhinagar · **Mar 2025 to Jul 2025**

- Analyzed lipid residue data from archaeological pottery (GC-MS lipid profiles and compound-specific carbon isotope ratios).
- Wrote Python pipelines to clean and compile the data into one database, compute summary statistics, and produce figures for the thesis: heatmaps, box plots and isotope biplots.

Senior Research Fellow (Dr. Vibin Ramakrishnan) · IIT Guwahati · **Jun 2022 to Sep 2022**

- Proteomics, cheminformatics, and network medicine analysis of Ashwagandha and Yograj Guggulu for osteoarthritis: protein-protein interaction analysis and computational drug repurposing.

M.Tech Thesis (Dr. Kusum K. Singh) · IIT Guwahati · **2021 to May 2022**

- RNA-seq identification and functional-enrichment analysis of nonsense-mediated-decay-susceptible transcripts in UPF3B-knockout HEK293 cells.

Publications

*Name in bold; * equal contribution.*

Peer-reviewed

1. Rakheja I.*, **Bharti V.***, *et al.* (2025) "Development of an *in silico* platform (TRIPinRNA) for the identification of novel RNA intramolecular triple helices and their validation using biophysical techniques." ***Biochemistry*** (ACS) 64(1):250-265. https://doi.org/10.1021/acs.biochem.4c00334 (**Equal contribution.*)
2. Sharma S., **Bharti V.**, *et al.* (2026) "*MLC1* alteration in human iPSCs give rise to disease-like cellular vacuolation phenotype in the astrocyte lineage." ***Orphanet Journal of Rare Diseases*** 21:176. https://doi.org/10.1186/s13023-026-04316-3 (*Open access.*)
3. Rakheja I., **Bharti V.**, Singh V., Maiti S. (2026) "Quercetin affects carcinogenic phenotype of breast and lung cancer cells differentially through sterol regulation mediated by MALAT1 perturbation." ***Chemistry: An Asian Journal*** 21(11):e70821. https://doi.org/10.1002/asia.70821 (*Open access.*)
4. Das P.K., Aich M., Adu P., **Bharti V.**, Maiti S., Chakraborty D. (2026) "CRISPR-Cas diagnostics (CRISPR-Dx) of viral pathogens in low- and limited-resource areas: current state and future directions." ***TrAC Trends in Analytical Chemistry*** 195:118590. https://doi.org/10.1016/j.trac.2025.118590
5. Rana P., Ujjainiya R., **Bharti V.**, Maiti S., Ekka M.K. (2024) "IGF2BP1-mediated regulation of *CCN1* expression by specific binding to a G-quadruplex structure in its 3′ UTR." ***Biochemistry*** (ACS) 63(17):2166-2182. https://doi.org/10.1021/acs.biochem.4c00172

Preprints and manuscripts

6. **Bharti V.**, Chakraborty D. (2026) "StickForStats: automated statistical assumption validation for reproducible computational biology." *bioRxiv*. https://doi.org/10.64898/2026.06.15.732278 (*First author and corresponding; under review,* BMC Bioinformatics.)
7. **Bharti V.**, Chakraborty D. (2026) "Most biomedical articles print no statistical result checkable by recomputation: a pre-registered audit with a 20,000-article out-of-sample replication." (*Manuscript in preparation; first and corresponding author.*)
8. Rauthan R., **Bharti V.**, *et al.* "An interface of genetically engineered human forebrain assembloids and polymeric nanofiber scaffolds for multiscale profiling of interneuron-migration disorders." *Research Square* preprint. https://doi.org/10.21203/rs.3.rs-3831019/v1 (*Under revision,* Stem Cell Reports.)
9. Kochar M., Rao S., Bhattacharjee S., Goel P., Timsina H., Azhar M.K., **Bharti V.**, *et al.* "Folding on the way in: vectorial substrate recognition by human Hsp60." (*Submitted to* Cell*, 2026.*)
10. Azam T., Kumar A., Singh P., **Bharti V.**, Ekka M.K. "Structural and functional analysis of the Chast-NUCB1 interaction in cardiac hypertrophy." (*Submitted to* Journal of Biological Chemistry*, 2026.*)

Research Software & Computational Methods

StickForStats: statistical-analysis platform with automatic assumption validation. First-author, open-source platform whose *Guardian* engine validates the assumptions behind every test (normality, variance homogeneity, independence, outliers) before it runs, then either reports with full confidence or cascades to the correct nonparametric alternative, with a companion module that re-checks the statistical claims in uploaded manuscripts. Built to counter the assumption violations that drive irreproducible results. *Best Poster Award, EMBO Conference 2025; first-author preprint on* bioRxiv*, under review at* BMC Bioinformatics*; public beta live at* stickforstats.com.

RNA Lab Navigator: retrieval-augmented research assistant. A retrieval-augmented LLM system over internal lab documents and the primary literature for fast, grounded literature triage in our lab, used by about 21 researchers.

Conferences, Talks & Posters

- Poster (selected for presentation), *Emerging Science, Technology, and Innovation Conclave (ESTIC) 2026*, Bharat Mandapam, New Delhi, 27 to 29 Oct 2026.
- **Best Poster Award**, "StickForStats: Publish Research with Confidence," **EMBO Conference 2025**, CCMB Hyderabad.
- Participant, *43rd Mahabaleshwar Seminar: Molecules, Membranes and Organelles* (TIFR), Alibaug, Maharashtra, 26 Feb to 1 Mar 2025; built and presented a simulation of organelles interacting with motor proteins under varying levels of molecular crowding in a breakout session.
- Poster, "Boundary Without Fence," *Frontiers in Genome Engineering 2023*, BITS Pilani (Goa).

Awards & Honors

- **Best Poster Award**, EMBO Conference 2025 (CCMB Hyderabad).
- **GATE Biotechnology: All-India Rank 127 (top ~1%)**, 2020.
- Graduate Assistantship, Ministry of Education, Government of India (2020 to 2022).
- GATE Electronics & Communication: top ~3 percentile, 2018.

Technical Skills

- **Languages & statistics:** Python, R, C, MATLAB; statistical modeling, experimental design, and test-selection methodology.
- **Genomics & bioinformatics:** scRNA-seq, RNA-seq, ChIP-seq; molecular docking, PyMOL; network analysis (Cytoscape, STRING, Gene Ontology).
- **ML & computational methods:** deep learning; retrieval-augmented LLM systems, vector databases, knowledge graphs; biomedical NLP (scispaCy, BioBERT).
- **Workflow:** Git, reproducible pipelines, Streamlit; scientific writing.

Selected Computational Projects

- **Deep learning for genome annotation.** Predicted DNA N6-methyladenine (6mA) sites in the rice genome with CNN and LSTM architectures; benchmarked predictive accuracy across models.
- **Sequence classification with RNNs.** Built an LSTM classifier to label DNA sequences by intronic, exonic, and exon-intron boundary context, improving over a CNN baseline.
- **Computational-biology collaboration (UC Berkeley).** Worked with a postdoctoral researcher at UC Berkeley to automate biological data-analysis pipelines in R and Python.

Teaching & Service

- **Teaching Assistant**, IIT Guwahati (Jun 2020 to May 2022): supported a class of 300+ students.
- **Core Organizing Team (IT lead)**, *Frontiers in Genome Engineering 2023*, BITS Pilani (Goa).
- **Mentor**, "One Week One Lab" outreach program, CSIR-IGIB (2023).
- **Chairperson**, IEEE Circuits & Systems Society Student Branch Chapter, IEM Kolkata (2016 to 2018).